Weekly Quiz 1

Instructor Key

Divisibility and the Euclidean algorithm

20 minutes • 10 points • Show the essential steps.

1. Use the Euclidean algorithm to compute $\gcd(252, 198)$. **[3 points]**

Solution

$$252 = 198 + 54, \quad 198 = 3 \cdot 54 + 36, \quad 54 = 36 + 18, \quad 36 = 2 \cdot 18.$$

The last nonzero remainder is 18, so $\gcd(252, 198) = 18$.

2. Explain why $\gcd(a, b) = \gcd(b, a - b)$ for positive integers $a > b$. **[3 points]**

Solution

A number d divides both a and b exactly when it divides both b and $a - b$: in one direction, $d \mid a$ and $d \mid b$ imply $d \mid a - b$; in the other, $d \mid b$ and $d \mid a - b$ imply $d \mid (a - b) + b = a$. Thus the two pairs have the same common divisors and therefore the same greatest common divisor.

3. A 252 cm $\times$ 198 cm rectangle is tiled by the largest possible congruent squares, with no cutting. Find the side length and the number of squares. **[4 points]**

Solution

The side length must divide both dimensions, so the largest possible side is $\gcd(252, 198) = 18$ cm. There are $252/18 = 14$ squares across and $198/18 = 11$ down, hence $14 \cdot 11 = 154$ squares.

Weekly Quiz 2

Instructor Key

Bézout identities and Diophantine equations

20 minutes • 10 points • Show the essential steps.

1. Find integers x, y such that $527x + 341y = \gcd(527, 341)$. **[4 points]**

Solution

Euclid gives

$$527 = 341 + 186, \quad 341 = 186 + 155, \quad 186 = 155 + 31, \quad 155 = 5 \cdot 31.$$

Back-substitution gives

$$31 = 186 - 155 = 2 \cdot 186 - 341 = 2 \cdot 527 - 3 \cdot 341.$$

Thus $\gcd(527, 341) = 31$ and one answer is $(x, y) = (2, -3)$.

2. Solve $35x + 22y = 1$ in integers, and give all solutions. **[3 points]**

Solution

Back-substitution in the Euclidean algorithm gives $1 = -5 \cdot 35 + 8 \cdot 22$, so one solution is $(-5, 8)$. Since $\gcd(35, 22) = 1$, all solutions are

$$x = -5 + 22t, \qquad y = 8 - 35t, \qquad t \in \mathbb{Z}.$$

3. Decide whether $42x + 30y = 18$ has integer solutions. If it does, find one. **[3 points]**

Solution

Since $\gcd(42, 30) = 6$ and $6 \mid 18$, solutions exist. Moreover, $6 = 3 \cdot 30 - 2 \cdot 42$, so $18 = 9 \cdot 30 - 6 \cdot 42$. One solution is $(x, y) = (-6, 9)$.

Weekly Quiz 3

Instructor Key

Primes and unique factorization

20 minutes • 10 points • Show the essential steps.

1. Factor 756 and 630 into primes; then find their gcd and lcm. **[4 points]**

Solution

$$756 = 2^2 \cdot 3^3 \cdot 7, \qquad 630 = 2 \cdot 3^2 \cdot 5 \cdot 7.$$

Taking the smaller exponent of each prime gives $\gcd(756, 630) = 2 \cdot 3^2 \cdot 7 = 126$. Taking the larger exponent gives $\operatorname{lcm}(756, 630) = 2^2 \cdot 3^3 \cdot 5 \cdot 7 = 3780$.

2. Let p be prime. Use Euclid's lemma to show that $p \mid a^2$ implies $p \mid a$. **[3 points]**

Solution

We have $a^2 = a \cdot a$. Euclid's lemma says that if a prime divides a product, it divides at least one factor. Both factors here are a, so $p \mid a$.

3. The statement "if $p \mid ab$, then $p \mid a$ and $p \mid b$" is false. Give a counterexample and state the correct theorem. **[3 points]**

Solution

For example, $2 \mid 2 \cdot 3$, but $2 \nmid 3$. The correct statement is Euclid's lemma: if p is prime and $p \mid ab$, then $p \mid a$ *or* $p \mid b$ (possibly both).

Weekly Quiz 4

Instructor Key

Continued fractions and best approximations

20 minutes • 10 points • Show the essential steps.

1. Evaluate the finite continued fraction $[3; 2, 4]$. **[3 points]**

Solution

$$[3; 2, 4] = 3 + \frac{1}{2 + \frac{1}{4}} = 3 + \frac{4}{9} = \frac{31}{9}.$$

2. Use Euclid's algorithm to find the simple continued fraction of $43/13$, and list its convergents. **[4 points]**

Solution

Since

$$43 = 3 \cdot 13 + 4, \qquad 13 = 3 \cdot 4 + 1, \qquad 4 = 4 \cdot 1,$$

we get $43/13 = [3; 3, 4]$. Its convergents are

$$3, \qquad 3 + \frac{1}{3} = \frac{10}{3}, \qquad [3; 3, 4] = \frac{43}{13}.$$

3. The first four convergents of $\sqrt{2} = [1; 2, 2, \ldots]$ are $1, 3/2, 7/5, 17/12$. Verify the Pell-type identity for the last convergent and explain what it says about the approximation. **[3 points]**

Solution

$$17^2 - 2 \cdot 12^2 = 289 - 288 = 1.$$

Thus $(17/12)^2 - 2 = 1/144$, so $17/12$ is extremely close to $\sqrt{2}$; equivalently, the numerator and denominator nearly solve $x^2 = 2y^2$.

Weekly Quiz 5

Instructor Key

Modular arithmetic

20 minutes • 10 points • Show the essential steps.

1. Find the least nonnegative residue of -137 modulo 12. **[2 points]**

Solution

Since $-137 = -12 \cdot 12 + 7$, we have $-137 \equiv 7 \pmod{12}$.

2. Solve $7x \equiv 5 \pmod{12}$. **[3 points]**

Solution

The inverse of 7 modulo 12 is 7, because $7^2 = 49 \equiv 1 \pmod{12}$. Multiplying by 7 gives $x \equiv 35 \equiv 11 \pmod{12}$.

3. Find the inverse of 17 modulo 43 using the Euclidean algorithm. Check your answer. **[5 points]**

Solution

$$43 = 2 \cdot 17 + 9, \quad 17 = 9 + 8, \quad 9 = 8 + 1.$$

Hence

$$1 = 9 - 8 = 2 \cdot 9 - 17 = 2(43 - 2 \cdot 17) - 17 = 2 \cdot 43 - 5 \cdot 17.$$

Therefore $17^{-1} \equiv -5 \equiv 38 \pmod{43}$. Indeed, $17 \cdot 38 = 646 = 15 \cdot 43 + 1$.

Weekly Quiz 6

Instructor Key

The Chinese remainder theorem

20 minutes • 10 points • Show the essential steps.

1. Solve $x \equiv 2 \pmod 3$ and $x \equiv 3 \pmod 5$. Give the complete congruence class. **[3 points]**

Solution

The numbers congruent to 3 modulo 5 are $3, 8, 13, \ldots$; among them, $8 \equiv 2 \pmod 3$. Thus $x \equiv 8 \pmod{15}$.

2. Solve $x \equiv 1 \pmod 4$ and $x \equiv 3 \pmod 7$. **[4 points]**

Solution

Write $x = 1 + 4k$. Then $1 + 4k \equiv 3 \pmod 7$, so $4k \equiv 2 \pmod 7$. Since $4^{-1} \equiv 2 \pmod 7$, $k \equiv 4 \pmod 7$. Therefore $x = 1 + 4(4) = 17 \pmod{28}$.

3. Explain immediately why $x \equiv 2 \pmod 6$ and $x \equiv 4 \pmod 9$ has no solution. **[3 points]**

Solution

Reducing both congruences modulo $\gcd(6, 9) = 3$ gives contradictory demands: the first says $x \equiv 2 \pmod 3$, while the second says $x \equiv 1 \pmod 3$. Hence no integer can satisfy both.

Weekly Quiz 7

Instructor Key

Necklace coloring and Fermat's little theorem

20 minutes • 10 points • Show the essential steps.

1. Five beads arranged in a circle are colored with three labeled colors. How many nonconstant color words are there, and how many rotation orbits do they form? **[4 points]**

   Solution

   There are 3^5 color words and 3 constant words, so $3^5 - 3 = 240$ are nonconstant. Because 5 is prime, every nonconstant word has an orbit of size 5. Therefore there are $240/5 = 48$ nonconstant rotation orbits.

2. Why does every nonconstant word of prime length p have exactly p distinct rotations? **[3 points]**

   Solution

   The orbit size divides p, so it is either 1 or p. Orbit size 1 means rotation by one place fixes the word, forcing every bead to have the same color. A nonconstant word cannot have size 1, so its orbit has size p.

3. Turn the orbit count into a congruence, then use it to compute $2^{11} \pmod{11}$. **[3 points]**

   Solution

   With a colors, the $a^p - a$ nonconstant words split into orbits of size p; hence $p \mid (a^p - a)$, or $a^p \equiv a \pmod{p}$. Thus $2^{11} \equiv 2 \pmod{11}$.

Weekly Quiz 8

Instructor Key

Groups and the standard proof of Fermat's theorem

20 minutes • 10 points • Show the essential steps.

1. List the units modulo 10 and the multiplicative inverse of each. **[3 points]**

Solution

The units are $\{1, 3, 7, 9\}$. Their inverses modulo 10 are $1^{-1} = 1$, $3^{-1} = 7$, $7^{-1} = 3$, and $9^{-1} = 9$.

2. List, in order, the elements generated by 3 in the multiplicative group modulo 7. What is the order of 3? **[3 points]**

Solution

The successive powers are

$$3^0, 3^1, \ldots, 3^5 \equiv 1, 3, 2, 6, 4, 5 \pmod 7,$$

and $3^6 \equiv 1$. Thus 3 generates all six nonzero classes and has order 6.

3. Let $p \nmid a$. Prove $a^{p-1} \equiv 1 \pmod p$ by multiplying the nonzero residue classes by a. **[4 points]**

Solution

Multiplication by a permutes $1, 2, \ldots, p-1$ modulo p: if $ai \equiv aj$, cancellation gives $i \equiv j$. Therefore

$$a \cdot 2a \cdots (p-1)a \equiv 1 \cdot 2 \cdots (p-1) \pmod p.$$

Hence $a^{p-1}(p-1)! \equiv (p-1)! \pmod p$. No factor of $(p-1)!$ is divisible by p, so it can be canceled, yielding $a^{p-1} \equiv 1 \pmod p$.

Weekly Quiz 9

Instructor Key

Binomial coefficients and divisibility

20 minutes • 10 points • Show the essential steps.

1. Compute $\binom{8}{3}$ and explain its subset interpretation. **[3 points]**

Solution

$$\binom{8}{3} = \frac{8 \cdot 7 \cdot 6}{3 \cdot 2 \cdot 1} = 56.$$

It counts the 3-element subsets of an 8-element set.

2. If p is prime and $1 \le k \le p-1$, prove that $p \mid \binom{p}{k}$. **[4 points]**

Solution

From

$$k\binom{p}{k} = p\binom{p-1}{k-1},$$

the right side is divisible by p. Since $1 \le k < p$, we have $p \nmid k$; Euclid's lemma therefore implies $p \mid \binom{p}{k}$.

3. Find the exponent of 2 in $\binom{10}{4}$, either by factoring or by counting binary carries. **[3 points]**

Solution

$$\binom{10}{4} = 210 = 2 \cdot 3 \cdot 5 \cdot 7,$$

so the exponent is 1. Equivalently, adding $4 = 0100_2$ and $6 = 0110_2$ produces exactly one carry, agreeing with Kummer's theorem.

Weekly Quiz 10

Instructor Key

Central binomial coefficients and primes

20 minutes • 10 points • Show the essential steps.

1. For $n = 4$, verify

$$\frac{4^n}{2n+1} \le \binom{2n}{n} \le 4^n.$$

[3 points]

Solution

Here $\binom{8}{4} = 70$, while $4^4/9 = 256/9 \approx 28.4$ and $4^4 = 256$. Thus $256/9 \le 70 \le 256$.

2. Show that every prime p with $n < p \le 2n$ divides $\binom{2n}{n}$. **[4 points]**

Solution

In

$$\binom{2n}{n} = \frac{(n+1)(n+2)\cdots(2n)}{n!},$$

the prime p appears in the numerator. Since $p > n$, it does not appear in $n!$. Hence the exponent of p in the binomial coefficient is positive, so $p \mid \binom{2n}{n}$.

3. Deduce the logarithmic estimate

$$\sum_{n<p\le 2n} \log p \le 2n \log 2,$$

and say in one sentence why this connects binomial coefficients to prime counts. **[3 points]**

Solution

The previous result shows $\prod_{n<p\le 2n} p \le \binom{2n}{n}$. Also $\binom{2n}{n} \le 4^n$. Taking logarithms gives

$$\sum_{n<p\le 2n} \log p \le \log \binom{2n}{n} \le n \log 4 = 2n \log 2.$$

The product records primes in an interval, while the central binomial coefficient gives a tractable size bound for that product.